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A neural-symbolic AI agent system for biomedical concept mapping



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Concept mapping links free-text descriptions to standardized concepts in biomedical terminologies and ontologies. Existing approaches fall into two main categories: rule-based systems, which are interpretable but struggle with ambiguity and scalability, and learning-based methods, which leverage contextual signals but perform poorly on long-tail concepts due to limited training data and offer limited explainability. To address these limitations, we propose Medical Concept Mapping (MCM), a novel agentic workflow that uses language models to rephrase ambiguous mentions into explicit, standardized descriptions prior to concept linking. This reformulation substantially improves mapping accuracy for underrepresented and abbreviated concepts. Across multiple benchmarks, MedMentions, ST21pv, and MCN, MCM outperforms state-of-the-art baselines, including KrissBERT, SciSpaCy, and USAGI, achieving Recall@1 scores of 63.3, 60.0, and 67.9, respectively. On zero-shot abbreviated mentions, MCM maintains strong performance, exceeding baseline Recall@1 by up to 24.8 points. Human evaluation confirms the quality of LLM-generated expansions: 79.4% were rated as reasonable and useful, and GPT-OSS received the highest approval (85.1%). These results demonstrate that MCM enables more accurate, interpretable, and robust long-tail concept normalization for biomedical applications.

Rapid expansion of biomedical literature and clinical documentation has led to an overwhelming volume of unstructured natural language data. Despite its richness, this data poses significant challenges due to the inherent ambiguity of biomedical or clinical language^{1–3}. Such ambiguity is particularly evident for named entities such as diseases, drugs, genes, and procedures, which frequently exhibit synonymy, abbreviation, and context-dependent variability. For example, “myocardial infarction” may be referred to as “MI” or “heart attack” and “RA” could refer to “rheumatoid arthritis”, “right atrium”, or “renal artery”.

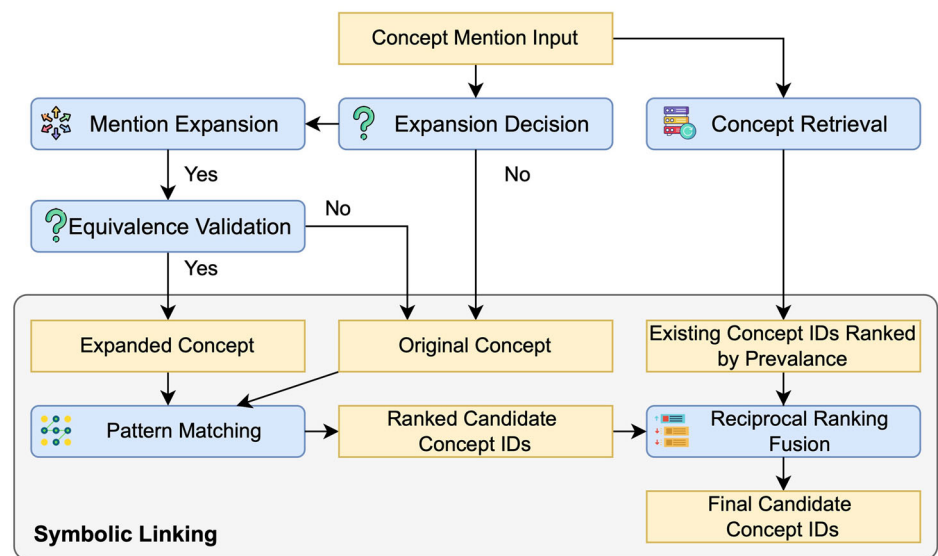
In parallel, the development of large-scale, high-quality medical terminologies, such as Systematized Nomenclature of Medicine - Clinical Terms (SNOMED CT)⁴, Medical Subject Headings⁵, and the Human Phenotype Ontology⁶, has enabled the structured representation of millions of medical concepts and their semantic relationships. Establishing links between unstructured biomedical text and these standardized terminologies is therefore essential for advancing text understanding, improving interoperability, and enriching existing knowledge resources. This task, commonly referred to as concept mapping, entity linking, or named entity normalization, serves as a cornerstone of biomedical text processing and

underpins a wide range of downstream applications, such as clinical question answering, relation extraction, information retrieval, and knowledge graph population^{7–20}.

Approaches to concept mapping can be broadly categorized into two major classes: (1) rule-based approaches and (2) learning-based approaches. Rule-based systems, which rely on handcrafted lexicons and heuristic rules, often exhibit high precision within narrowly defined domains but struggle to generalize across diverse and linguistically variable texts^{3,21,22}. Learning-based approaches, on the other hand, improve adaptability by leveraging statistical models that incorporate engineered lexical, syntactic, and semantic features to handle ambiguity and previously unseen mentions^{23–26}. More recently, deep learning methods have used distributed and contextual embeddings^{27–29} to capture fine-grained semantic nuances in biomedical text^{30–33}. However, these models often underperform on concepts that are underrepresented in training data, limiting their applicability in long-tail or emerging domains. Large language models (LLMs) have also been used to produce single-sentence descriptions that explain concepts. The explanations are then encoded to align with standardized terminologies³⁴. However, their effectiveness is limited in domains where reference definitions are

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Fig. 1 | The overview of MCM, an AI agent that links concepts to terminology entries via dynamically expanding concept mentions and integrating pattern-matching results.



already complete, and their general-domain pretraining may fail to capture the fine-grained distinctions critical for biomedical tasks. Additionally, the opaque, black-box nature of deep learning poses challenges for transparency, interpretability, and error analysis, thereby hindering trust and quality assurance in clinical and research applications.

Building on these observations, we propose a neural-symbolic agent system, MCM (Medical Concept Mapping), which integrates the complementary strengths of neural and symbolic paradigms: i.e., neural models provide robust pattern recognition and adaptability, while symbolic reasoning adds interpretability and consistency (Fig. 1). Unlike prior LLM-assisted entity linking methods that apply prompt-based expansion or generation in a single step, MCM decomposes the task into explicit stages that conditionally invoke neural reasoning only when needed. The system first determines whether a mention requires refinement, then generates a context-aware expansion, and subsequently validates semantic equivalence before symbolic linking. This design enables controlled use of LLMs while preserving deterministic behavior through symbolic matching. For concepts well-represented in the training data, MCM dynamically reuses existing annotations, whereas for rare or abbreviated mentions, it activates the expansion and validation modules. By explicitly separating decision, expansion, validation, and linking, MCM provides a transparent and extensible modular framework that improves long-tail concept mapping while maintaining interpretability. We evaluated MCM against state-of-the-art methods across three benchmark datasets. We showed that MCM consistently outperforms KrissBERT, ScispaCy, and USAGI, achieving state-of-the-art Recall@1 scores on these benchmark datasets and showing up to 24.8% gains on zero-shot abbreviated mentions. Our analysis further showed that performance is influenced by model architecture and context window size, with longer contexts enabling LLMs to infer concept meanings even when explicit expansions are absent.

Results

Overview of the study

MCM is designed to improve both the accuracy and interpretability of concept mapping by integrating retrieval, refinement, and validation into a unified workflow (Fig. 1). We tested MCM on three publicly available concept mapping benchmarks, derived from clinical notes and scientific abstracts, representing realistic and heterogeneous biomedical text sources. Specifically, MedMentions and ST21pv are derived from PubMed abstracts³⁵, while the n2c2 2019 Medical Concept Normalization (MCN) dataset is derived from MIMIC clinical notes⁵. Each mention in these datasets is manually annotated with a corresponding UMLS Concept

Unique Identifier (CUI), which serves as the gold standard for concept mapping. We adopted the commonly used Recall@K metric, which measures the proportion of test mentions for which the correct CUI is found in the top-K outputs. Experiments were conducted using multiple open-source models as backbones, including LLaMA 3.1 (8B)³⁶, GPT-OSS (20B)³⁷, Qwen-3 (32B)³⁸, DeepSeek-R1³⁹, Phi-4⁴⁰, Gemma-2⁴¹, and Mistral (7B)⁴². For all metrics, 95% confidence intervals (95% CI) were computed using a bootstrap resampling procedure with 1000 iterations. Statistical significance was determined using paired *t*-tests with $p < 0.01$. Further methodological details are provided in the Methods section.

MCM agent improves the concept mapping across three datasets

We first evaluated MCM against three state-of-the-art baselines: USAGI²⁶, ScispaCy²², and KrissBERT²⁷, across three benchmarks. As shown in Fig. 2, MCM consistently outperforms all baselines. On MedMentions, MCM achieves a Recall@1 of 63.3, outperforming KrissBERT (58.0), ScispaCy (55.9), and USAGI (30.7) by 5.3, 7.4, and 32.6 points, respectively. On ST21pv, MCM reaches a Recall@1 of 60.0, surpassing KrissBERT (54.6), ScispaCy (53.7), and USAGI (36.4) by 5.4, 6.3, and 23.6 points. On MCN, it scores 67.9, leading KrissBERT (63.1), ScispaCy (60.1), and USAGI (53.1) by 4.8, 7.8, and 14.8 points, respectively.

To determine the optimal backbone LLM for MCM, we compared the concept linking performance across 7 different LLMs (Supplementary Fig. 1). The top-performing models were GPT-OSS and Gemma 2, achieving Recall@1 scores of 38.4 and 37.5, respectively. Phi-4 and LLaMa 3.1 followed closely, with Recall@1 scores of 36.2 and 36.0. These models outperformed others, while Qwen3 lagged behind with a Recall@1 of 33.1, indicating relatively weaker performance in retrieving the correct concept as the top prediction. Based on these results, GPT-OSS was selected as the backbone model for MCM.

Zero-shot mapping of abbreviated mentions

We next evaluated MCM's performance on the most challenging zero-shot abbreviated cases to assess its ability to generalize to unseen and shortened concept mentions. Figure 3 presents MCM's Recall@1 performance on these cases—concept mentions that are both unseen during training and abbreviated—across the three datasets. MCM achieves Recall@1 scores of 36.3 on MedMentions, 36.7 on ST21pv, and 38.6 on MCN, consistently outperforming all baseline systems. Specifically, it surpasses KrissBERT (11.5, 12.0, 28.8) by 24.8, 24.7, and 9.8 percentage points, ScispaCy (20.7, 20.5, 33.3) by 15.6, 16.2, and 5.3 points, and USAGI (6.1, 6.0, 21.6) by 30.2,

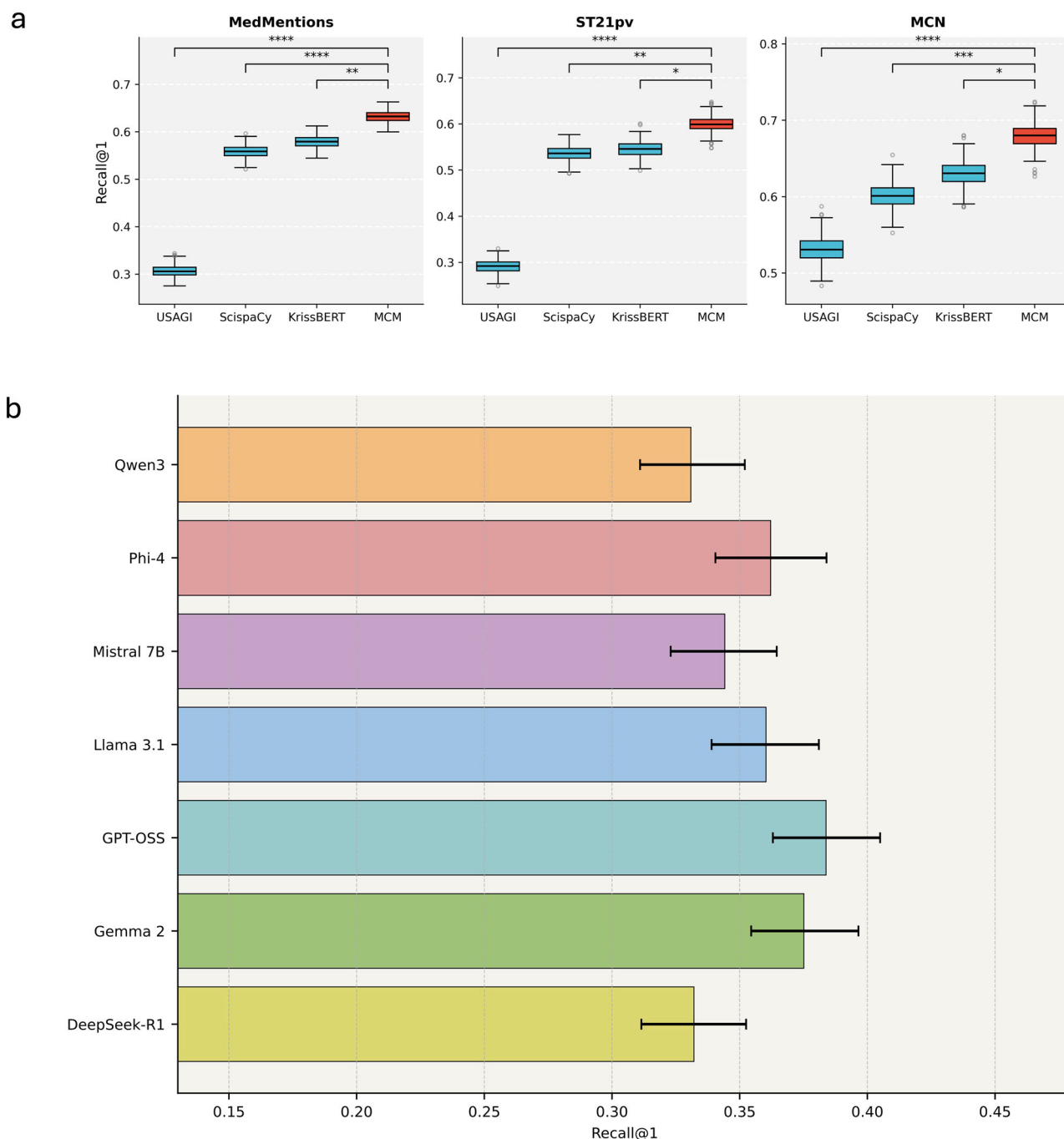


Fig. 2 | Performance of MCM and backbone LLMs across benchmarks. a The performance of our proposed MCM and three baselines on MedMentions, ST21pv, and MCN. * $p > 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns: Not significant.

b Recall@1 across 7 LLMs serving as backbone components in the MCM, evaluated on a 2000-sample subset using a context window of 256 tokens. Error bars represent 95% confidence intervals based on 1000 bootstrap resamples.

30.7, and 17.0 points, respectively. These improvements are substantially greater than those observed in the overall evaluation (Fig. 2), highlighting MCM's robust generalization ability in the most challenging zero-shot and abbreviated scenarios. Notably, while KrissBERT generally outperforms ScispaCy in overall evaluations, it yields lower Recall@1 in these zero-shot cases, illustrating the limitations of learning-based models in long-tail or out-of-distribution settings.

Here we provide two examples of concept expansions. The first case involves the abbreviated mention “Py” in a chemistry-focused passage describing pyrrole–imidazole polyamides. Within the surrounding context “... design of polyamide 1 was based on a conventional model, with three consecutive Py or Im rings separated by a β -alanine to match the curvature

and twist of long DNA helices...”, “Py” is used to denote a specific chemical unit, N-methylpyrrole, which is explicitly introduced in the text. In this instance, the expansion module generates “pyrrole” as the expanded form. While this expansion is semantically related and broadly correct, it reflects a partial loss of specificity, as the methyl substitution indicated in the original context is not preserved. The second case concerns the abbreviation “PAR” in an electrochemical study on the simultaneous determination of pharmaceutical compounds. The context, “...was fabricated on the glassy carbon electrode (GR-NF/GCE), and used for simultaneous determination of paracetamol (PAR), aspirin (ASA) and caffeine (CAF)...”, explicitly defines “PAR” as paracetamol, alongside aspirin and caffeine. Here, the expansion module correctly resolves the abbreviation to paracetamol and further maps

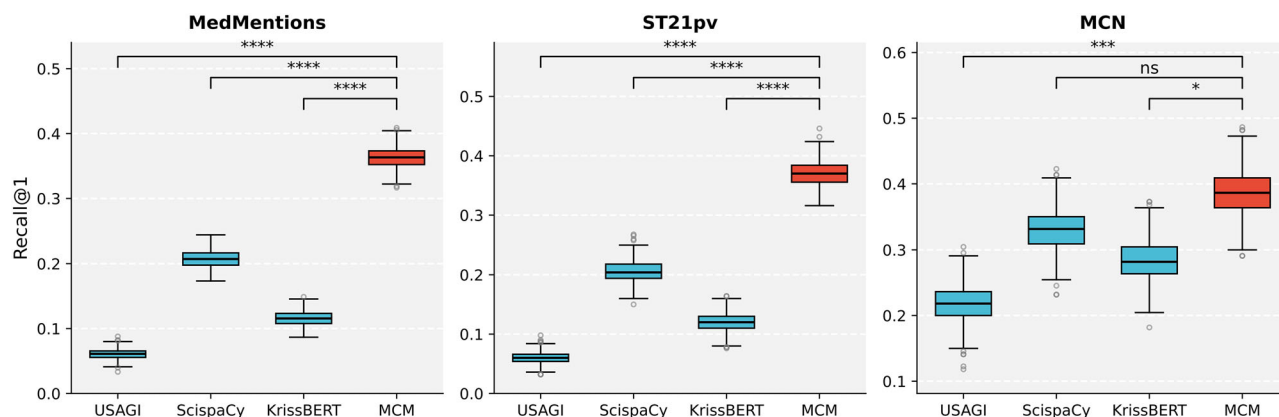


Fig. 3 | The performance of our proposed MCM and other baselines on zero-shot concepts. * $p > 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns Not significant.

it to its preferred name, maintaining both semantic accuracy and terminological consistency. This example demonstrates the model's ability to effectively exploit explicit definitional context to perform accurate abbreviation expansion and concept normalization when the abbreviation–full form relationship is clearly stated.

Performance breakdown across different semantic types

To assess the impact of mention expansion on concept mapping, we analyzed performance gains across different semantic types following symbolic linking. Figure 4 shows large absolute improvements for multiple semantic types: Organic Chemical increasing from 22.8 to 80.4 (+57.6), Pharmacologic Substance from 20.7 to 78.4 (+57.7), Neoplastic Process from 25.3 to 76.9 (+51.6), Therapeutic or Preventive Procedure from 5.5 to 50.4 (+44.9), Disease or Syndrome from 31.2 to 77.9 (+46.7), Element, Ion, or Isotope from 24.7 to 71.9 (+47.2), Body Part, Organ, or Organ Component from 36.6 to 70.9 (+34.3), and Cell from 23.6 to 59.6 (+36.0). Moderate gains are observed for Finding (+27.2), Immunologic Factor (+19.0), and Enzyme (+8.4), among others. The notable exception is Gene or Genome, where recall decreases from 67.1 to 47.5 (−19.6). Overall, once mentions are correctly expanded, the symbolic component resolves ambiguity effectively for most categories, delivering substantial improvements with only a few exceptions.

Expert evaluation of concept expansions

To complement the quantitative evaluations, we conducted a human evaluation study to assess the quality and usefulness of LLM-generated concept expansions for concept mapping. The majority of ratings reflected a positive assessment of the LLM-generated expansions (Fig. 5, and Supplementary Table 1). Specifically, 69.8% of the ratings were “Strongly Agree” and 9.5% were “Agree,” indicating that approximately 79.4% of the evaluated expansions were perceived as reasonable, meaningful, and useful for concept mapping. Inter-rater reliability, measured by a quadratic weighted Cohen's kappa, was 0.342, with an overall agreement rate of 55.3% and a relaxed agreement rate of 77.1% (defined as a difference of no more than one rating point). Neutral responses accounted for 5.6%, while 10.5% and 4.5% of responses were “Disagree” and “Strongly Disagree,” respectively. Among the models evaluated, GPT-OSS received the most favorable ratings, with 77.7% as “Strongly Agree” and 7.4% as “Agree” (85.1% positive). Gemma 2 (81.9% positive) and LLaMA 3.1 (81.5%) followed closely, while Phi-4 also performed strongly with 80.8% positive ratings. Mistral 7B (76.1%), Qwen-3 (75.5%), and DeepSeek-R1 (74.5%) also demonstrated solid performance.

Impact of context length on accuracy and efficiency

We further investigated how the length of input context affects concept mapping performance, testing whether longer context windows improve LLMs' ability to resolve ambiguous or abbreviated mentions. Across all evaluated models, recall consistently improved as the context length

increased. For example, GPT-OSS increased from 36.0 at 32 tokens to 44.0 at 256 for Recall@1 (Fig. 6a). Gemma-2 showed steady gains, rising from 34.0 to 44.0 as the context expanded to 256 tokens. DeepSeek-R1 improved from 32.0 at length 32 tokens to 40.0 at 256, while Qwen3 demonstrated the largest relative improvement, increasing from 20.0 to 36.0. Overall, all models benefited from longer contexts, with lower-performing models (e.g., Mistral, Qwen-3) showing greater gains from the richer context information. Among all evaluated models, GPT-OSS, Phi-4, and Gemma-2 consistently achieved the highest recalls across different context window settings, reaching approximately 44.0, while DeepSeek-R1 and Qwen3 remained comparatively lower.

To determine whether longer context windows incur computational costs, we analyzed model runtimes across varying context lengths. To evaluate this, we measured runtime across models at different context lengths using Nvidia L40S (Fig. 6b). The results show that extending the context window from 32 to 256 tokens did not lead to statistically significant increases in processing time. For example, DeepSeek-R1 and Gemma-2 maintained nearly identical runtimes across all context window sizes, while Mistral and Qwen-3 exhibited only minor fluctuations without a consistent upward trend. These findings suggest that larger context windows can be adopted to improve recall without incurring meaningful computational penalties.

Supplementary Table 2 reports an ablation analysis evaluating the contribution of the expansion and checking modules across three benchmarks (MedMentions, ST21pv, and MCN) using Recall@1 and Recall@5. Removing the expansion module results in a consistent performance drop across all datasets; notably, this setting is equivalent to the SciSpacy baseline, where mentions are mapped directly without expansion, with the largest degradation observed on MCN (Recall@1: from 0.679 to 0.601; Recall@5: from 0.832 to 0.748), indicating that concept expansion is particularly important for long-tail and ambiguous concept mentions. Without the checking module, expansion is applied less selectively, leading to suboptimal candidate generation and reduced recall. When both modules are enabled (MCM Full), the system achieves the best performance on all benchmarks.

Discussion

MCM demonstrates significant improvement over current state-of-the-art methods, with particularly strong improvements on zero-shot abbreviations not present in the training data (Fig. 2). This advantage stems from MCM's ability to leverage contextual information to expand ambiguous or under-specified concept mentions before linking (Fig. 3). We observe that performance improves steadily with longer context windows, indicating that richer local context supports more accurate expansion. This effect is particularly pronounced in MedMentions compared to MCN, which may reflect differences in domain-specific writing practices: scientific abstracts typically define abbreviations before use, whereas clinical notes often use abbreviations without explanation. Furthermore, we show that mapping accuracy

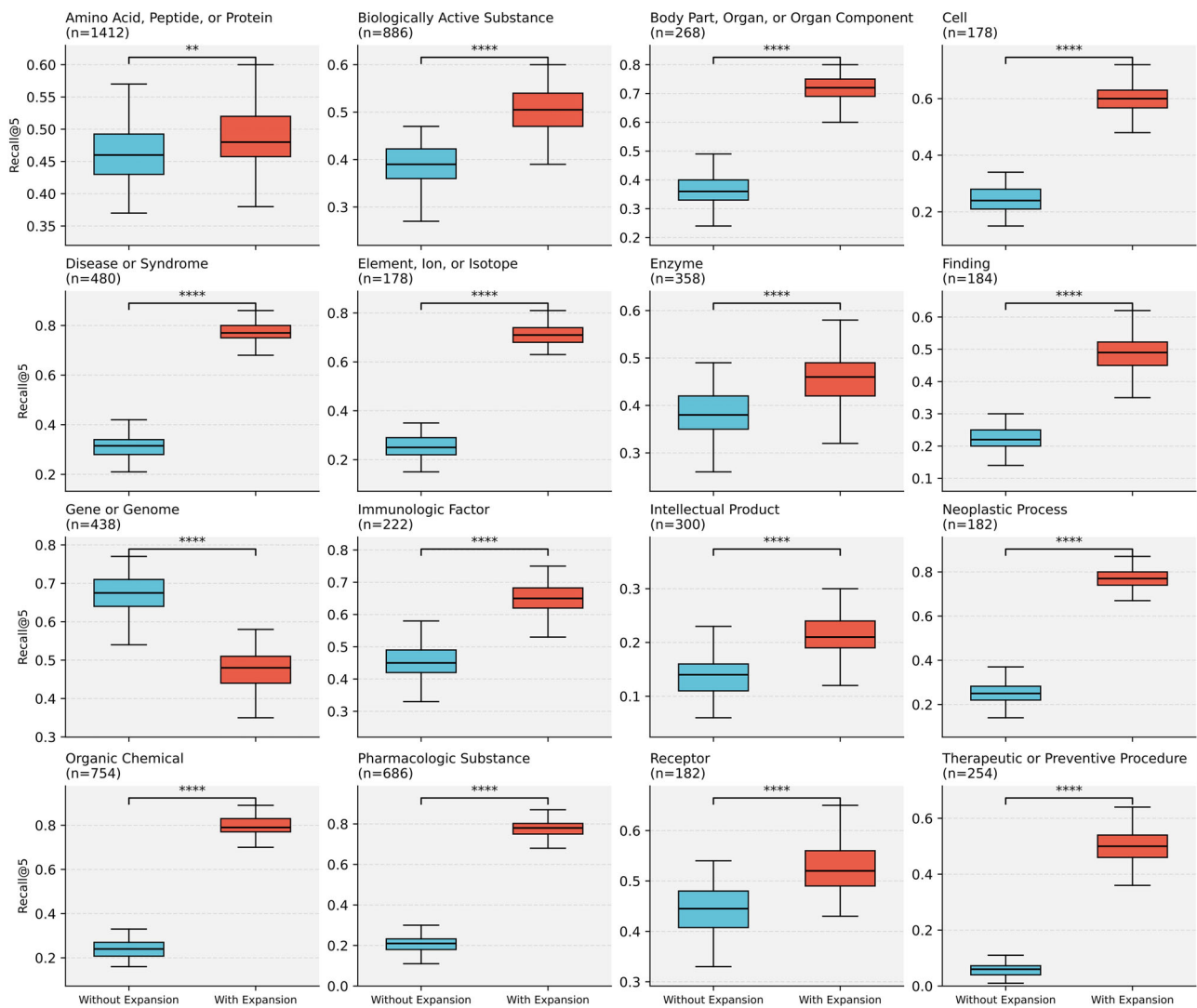
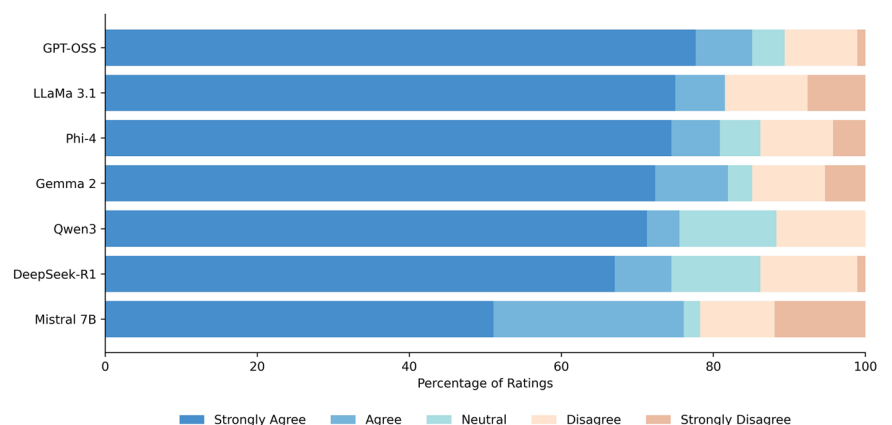


Fig. 4 | Concept mapping performance by semantic types. The left bar shows performance without expansion. The right bar shows performance with expansion. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns Not significant.

Fig. 5 | Distribution of human ratings for concept expansions generated by each model. Each horizontal bar represents the proportion of ratings (on a 5-point Likert scale) for concept expansions generated by a specific model. Ratings reflect the agreement of human evaluators that an expansion is reasonable, meaningful, and helpful for concept mapping. The five rating levels are color-coded from left to right: Strongly Agree (5), Agree (4), Neutral (3), Disagree (2), and Strongly Disagree (1).



significantly improves across most semantic categories after expansion (Fig. 4), confirming that disambiguating mentions before linking substantially enhances overall accuracy. Our findings also highlight the contribution of parametric knowledge encoded within LLMs, allowing them, in some cases, to infer the intended concept when the full form of an abbreviation is absent.

For example, the term “TCA” is correctly expanded to “Trichloroacetic Acid” in an abstract (PMID: 27379139), despite not being explicitly defined. However, the same term can also be interpreted as “Tricarboxylic Acid Cycle” or “Tricyclic Antidepressant,” illustrating that concept mapping based solely on parametric knowledge remains inconsistent and is further

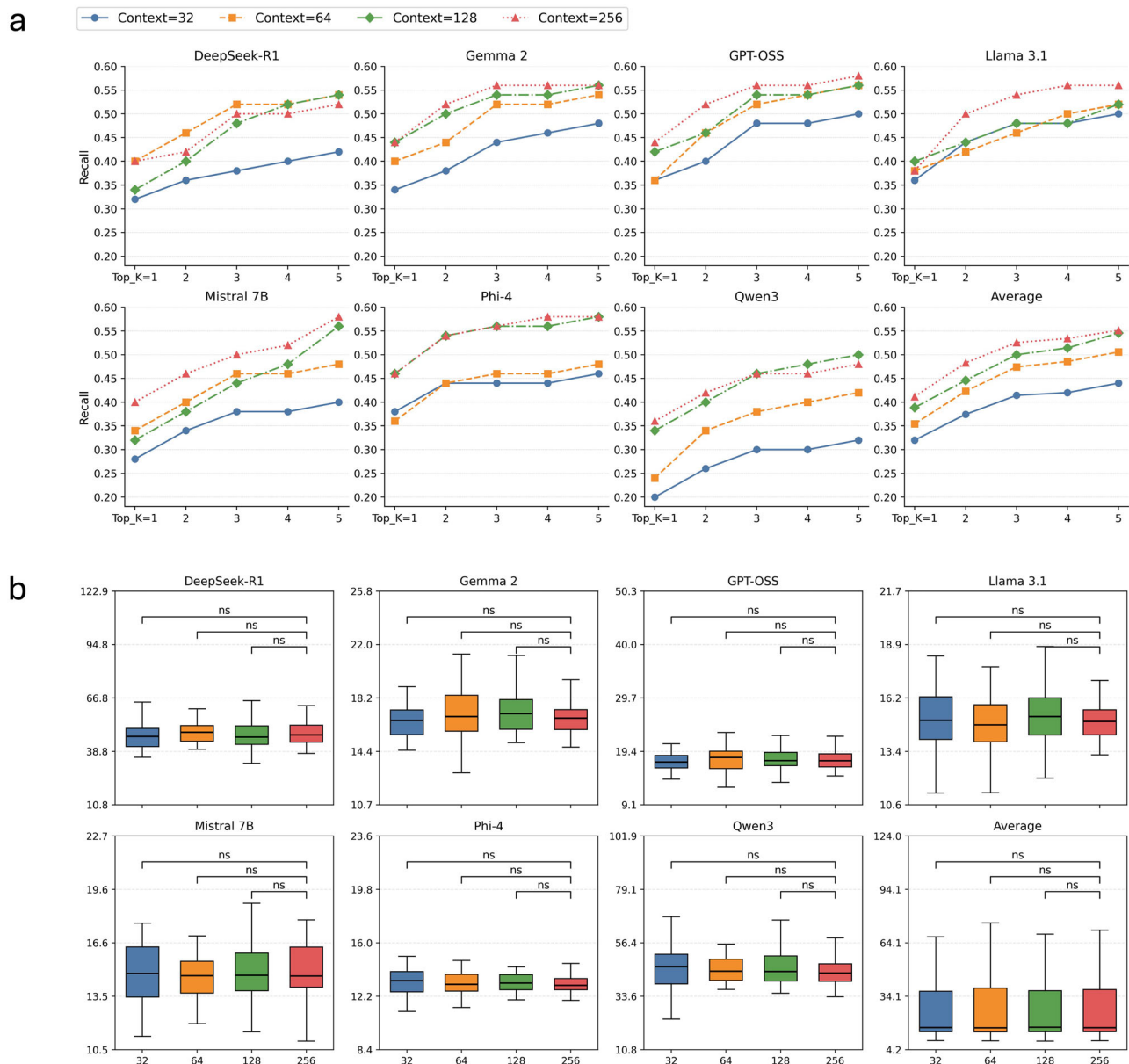


Fig. 6 | Impact of context window on end-to-end performance and runtime. a Recall from top 1 to 5 of MCM with DeepSeek-R1, Gemma2, GPT-OSS, Llama3.1, Mistral, Phi-4, and Qwen-3 as backbone. **b** The average runtime of MCM to process a concept mention. * $p > 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns Not significant.

limited for terminology entries without descriptive definitions. While MCM substantially improves zero-shot mapping accuracy compared to existing baselines, we note that absolute Recall@1 performance on unseen abbreviated mentions remains limited, reflecting the inherent difficulty of fully automating concept normalization in such cases. Importantly, MCM is flexibly designed to expose intermediate decision signals that can be used to identify potentially unreliable mappings. In particular, the Expansion Decision and Equivalence Validation modules provide explicit judgments on whether a mention can be meaningfully refined and whether the refined form preserves the original semantics in context. When equivalence validation fails, the system intentionally falls back to symbolic linking using the original mention, which may serve as an implicit indicator of low confidence in LLM-driven refinement.

One major challenge in biomedical concept mapping is the highly skewed, long-tail distribution of terminology, where a small portion of concepts appear frequently in training data, while the majority remain underrepresented. This imbalance leads to substantial performance gaps, particularly for rare terms. Our evaluation (Figs. 2 and 3) shows noticeable

disparities between well-represented and zero-shot concepts for both ScispaCy and KrissBERT, with KrissBERT exhibiting the largest drop in recall for infrequent mentions. These findings suggest that purely learning-based approaches, even those built on powerful pretrained models, struggle to generalize beyond their training distributions. In contrast, MCM achieves significant improvements on underrepresented concepts by explicitly expanding ambiguous or sparse surface forms and integrating symbolic linking strategies that do not rely on training frequency.

Another key advantage of our workflow is its interpretability. While recent learning-based models, such as KrissBERT, offer strong performance on many concepts, their errors are often opaque and difficult to explain. For instance, when mapping the mention “CJS” (Curry Jones syndrome, UMLS CUI: C0795915) from a PubMed abstract (PMID: 28386950), KrissBERT’s top predictions are Creutzfeldt-Jakob disease (UMLS CUI: C0022336) and Adenocarcinoma of prostate (UMLS CUI: C0007112), which are unrelated to the intended concept and appear to reflect latent embedding similarities that are not human-interpretable. In contrast, ScispaCy fails due to abbreviation ambiguity, returning Culler-Jones syndrome (UMLS CUI:

C4014479), which shares the same abbreviation as the target concept. Our workflow successfully addresses this challenge by first expanding “CJS” to “Curry Jones syndrome” and then linking it correctly through symbolic pattern matching. This stepwise process enables intermediate representations, enabling inspection of how decisions are made and thereby making the reasoning path far more transparent.

Our analysis across various semantic types shows that most categories benefit substantially from mention expansion, with particularly large improvements for diseases, chemicals, procedures, and anatomical entities (Fig. 4). However, the Gene or Genome category is a notable exception, where performance declines following expansion. This decrease in Recall@1 is largely attributable to gene–protein name string collisions, in which identical surface forms are used to denote both a gene and its corresponding protein (e.g., PLK1, CD79A), leading expanded mentions to retrieve closely related but semantically distinct concepts ranked above the gold standard. As a result, expansion can shift the top prediction across semantic types even when the correct concept remains highly ranked. Similarly, the Amino Acid, Peptide, or Protein category shows only minimal improvement. This pattern likely reflects the unique complexity of gene and protein nomenclature, which is highly aliased and sensitive to formatting conventions. Unlike other biomedical terms, gene and protein names typically follow standardized symbolic nomenclature established by official committees such as the HUGO Gene Nomenclature Committee. These names are usually alphanumeric and may represent both a gene and its corresponding protein product, introducing ambiguity without clear contextual cues. Because these preferred symbols are already consistently used in biomedical literature and terminology, such as UMLS, expanding them into full descriptions is unnecessary and can even impair mapping accuracy by introducing erroneous forms. A simple heuristic, such as skipping expansion when a mention can be reliably recognized as a gene symbol, may help avoid expanding gene or protein names. Robustly recognizing gene names requires dedicated named entity recognition (NER) and typing mechanisms, which are beyond the scope of the current work.

To address the heterogeneous nature of biomedical concept mentions, we implemented MCM as a multi-agent system that dynamically orchestrates its submodules based on the characteristics of each input. This design offers key advantages. First, it enables adaptive processing based on annotation availability: for frequent, well-represented mentions in training data, the workflow can directly retrieve nearest neighbors from the knowledge base; whereas for rare, abbreviated, or context-dependent terms, it automatically invokes the expansion and validation modules to refine the mention before linking. Second, MCM is designed to be extensible to related tasks such as NER. Our agent orchestrator (Supplementary Fig. 1) automatically distinguishes between three core functions: concept mapping, NER, and concept detail lookup, each treated as a separate task within the system.

Our study has several limitations. First, our evaluation focuses exclusively on isolated concept mapping rather than a full end-to-end pipeline from NER to concept mapping. In real-world applications, span detection errors from NER may propagate and negatively affect mapping accuracy. We leave this as a direction for future work. Second, the agent-based design of our workflow introduces additional computational overhead relative to purely rule-based or learning-based approaches, which could pose latency concerns in real-time settings. This issue can be substantially alleviated by caching frequently occurring terms and their mapping results. Finally, our system retrieves candidate concept IDs using a lexical-similarity approach for concepts that are frequently observed in the training data. While we did not deeply optimize this module, this design choice does not undermine our main conclusions, as our primary focus is on underrepresented and abbreviated mentions.

In summary, we introduce MCM, an advanced approach to biomedical concept mapping that combines neural mention expansion and symbolic

linking. This hybrid design significantly improves mapping accuracy, particularly for underrepresented and abbreviated terms, as demonstrated across multiple datasets and semantic types. We hope MCM will assist future efforts to more accurately extract, normalize, and represent scientific knowledge from textual sources.

Methods

MCM design and implementation

MCM comprises five modules designed to enhance the accuracy and robustness of biomedical concept mapping, which takes as input a biomedical mention and its surrounding context, and returns a ranked list of standardized UMLS concept identifiers (CUIs) that best represent its meaning. Among these five modules, the *Expansion Decision*, *Mention Expansion*, and *Equivalence Validation* modules are LLM-based and not specialized through additional training, while the *Concept Retrieval* and *Symbolic Linking* modules are symbolic and rely on precomputed lexical similarity indices (Fig. 1). First, the *Concept Retrieval* module searches for existing labeled mentions that are lexically similar to the input and retrieves their annotated concept identifiers, ranking candidate identifiers by prevalence to prioritize commonly observed matches if available. The *Expansion Decision* module then evaluates whether the input mention could be expanded or clarified to improve alignment with standardized terminology. If refinement is warranted, the *Mention Expansion* module generates a more complete or explicit form of the mention. The *Equivalence Validation* module subsequently determines whether the expanded mention preserves the same meaning as the original in its given context. Finally, the *Symbolic Linking* module applies symbolic pattern matching using either the original mention (if the expansion is not equivalent) or the expanded mention (if equivalent), and integrates the results with those from similar mention retrieval via reciprocal rank fusion to produce the final ranked list of concept identifiers.

The *Concept Retrieval* module identifies concept identifiers (CUIs) associated with previously annotated mentions that are lexically similar to the current input mention. To quantify this similarity, the input mention m is encoded using a character-level term frequency-inverse document frequency (TF-IDF) vectorizer, producing a sparse vector representation $v_m = \text{tfidf}(m) \in \mathbb{R}^d$. This vector is then compared against a precomputed matrix of TF-IDF vectors for annotated mentions $V_{\text{ann}} \in \mathbb{R}^{N \times d}$, where each row corresponds to an example mention from the labeled corpus. Pairwise cosine similarity is computed as $s = \cos(v_m, V_{\text{ann}})$, and all indices i such that $s_i \geq \tau_{\text{sim}}$ are retained, where τ_{sim} is a predefined similarity threshold. The CUIs associated with these matched examples are then extracted and aggregated, with a frequency-based ranking used to prioritize the CUIs that occur most frequently.

The *Expansion Decision* module serves as the first stage of refinement in the neural component of MCM, aiming to detect whether a biomedical mention requires clarification or rephrasing. Given an input mention m and its surrounding context c , the module determines whether m is expressed in an abbreviated, shorthand, or otherwise non-standard form. This decision is modeled as a binary function: $\text{NeedExpansion}(m, c) \rightarrow \text{bool}$ where the output is *True* if expansion is deemed necessary and *False* otherwise (Box 1). This decision is made by prompting the LLM backbone, asking it to assess the clarity and canonicity of the expression. Mentions flagged for expansion proceed to the next module in the workflow, while those deemed already clear continue unchanged.

The *Mention Expansion* module aims to improve the clarity of abbreviated biomedical mentions by generating a more explicit version. Given a mention m and its contextual snippet c , this module prompts the LLM backbone to rewrite m into a refined form m' that better aligns with canonical expressions used in biomedical terminologies (Box 2), denoted as: $\text{Augment}(m, c) \rightarrow m'$. The rewritten mention is then passed to the next module for quality assurance.

The *Equivalence Validation* module ensures that the augmented mention m' , generated in the previous step, faithfully retains the semantic meaning of the original mention m within its context c , via a self-feedback mechanism⁴³. The module invokes the backbone to perform a controlled comparison (prompt shown in Box 3), issuing a binary judgment on whether m' can reasonably be interpreted as semantically equivalent to m given c , denoted as $\text{Equivalent}(m, m', c) \rightarrow \text{bool}$. If the equivalence judgment returns *True*, the refined mention m' is propagated to the symbolic linking stage. If it is not judged equivalent, the system falls back to the original mention m .

The *Symbolic Linking* module performs deterministic lexical similarity-based mapping between the final mention m^f (either the original m or a refined version m') and standardized UMLS concepts. First, m^f is represented as a character-level TF-IDF vector $v_f = \text{tfidf}(m^f) \in \mathbb{R}^d$ and compared against a precomputed TF-IDF matrix of UMLS concept names $V_{\text{umls}} \in \mathbb{R}^{N_{\text{umls}} \times d}$. Cosine similarity is used to score each candidate concept i , yielding a similarity score $\text{similarity} = \cos(v_f, V_{\text{umls}}[i])$. The top k most similar concepts are retrieved. To integrate this output with previously retrieved candidates from the Concept Retrieval module, the system applies Reciprocal Rank Fusion (RRF). Given the two ranked lists of candidates, the final score for each candidate CUI is computed as $\sum_j \frac{1}{k_j + r_j(\text{CUI})}$, where r_j (CUI) is the rank of the CUI in the source j , and k_s is a smoothing constant.

MCM is implemented using several libraries: scikit-learn⁴⁴ for TF-IDF encoding and similarity calculations, LangChain⁴⁵ for orchestrating prompt templates and structured interactions with local LLMs, and Model Context Protocol⁴⁶ for managing modular, stateful communication between LLM-driven agents and MCM. We used $\tau_{\text{sim}} = 0.95$ and $k_s = 60$ in our implementation.

Box 1: | Determining whether a mention needs clarification or refinement

You are a biomedical language expert. Given the following concept mention, determine whether the mention is expressed as regular words.
 Concept mention: "{mention}"
 If the term is not a regular word, respond 'true'.
 Otherwise, respond 'false'.
 Respond in JSON format:
 {"result": true / false, "explanation": "brief justification"}

Box 2: | Clarifying biomedical concept mentions using context

You are a biomedical expert tasked with improving the clarity of a biomedical concept mention.
 Concept mention: "{mention}"
 Context: "{context_snippet}"
 If the mention is abbreviated or shortened, replace it with a fully spelled term mentioned in the context.
 If the mention is already standard and clear, keep it unchanged.
 Respond in JSON format:
 {
 "augmented_mention": "string",
 "explanation": "brief justification"
 }

Box 3: | Evaluating semantic equivalence between original and expanded mentions

You are a biomedical expert evaluating whether a refined phrase reasonably captures

the same meaning as the original concept mention, given the surrounding clinical context.

Original mention: "{mention}"
 Context: "{context_snippet}"
 Refined phrase: "{refined_phrase}"
 Only respond 'true' if the refined phrase can reasonably be interpreted to mean the same thing in context, even if there are minor wording differences.
 Only respond 'false' if the meaning is clearly different.
 Respond in JSON format:
 {
 "result": true / false,
 "explanation": "brief justification"
 }

Dataset and experiments

To develop our workflow and enable scalable evaluation, we utilized three benchmark datasets for biomedical concept normalization: MedMentions, ST21pv, and the n2c2 2019 MCN. MedMentions derived from PubMed abstracts and contains 352,496 annotated mentions spanning roughly 1% of the 3.2 million UMLS concepts. ST21pv is a subset of MedMentions, focusing on 203,282 mentions from 21 semantic types across preferred vocabularies. MCN is derived from MIMIC clinical notes and comprises 13,609 annotated mentions. It offers a complementary view of clinical language in practice.

We adopted Recall@K as the primary evaluation metric, which quantifies the proportion of test mentions for which the correct concept identifier appears within the top-k system outputs. To challenge our system with more difficult scenarios, we focused on long-tail concept mentions, specifically those that (1) do not appear in the training data and (2) are expressed in abbreviated or underspecified forms. This filtering yielded 2941 examples from MedMentions, 2512 from ST21pv, and 232 from MCN, forming a robust testbed for evaluating performance on rare, ambiguous, or unseen biomedical terminology.

We surveyed a wide range of existing concept mapping tools and selected representative, actively maintained baselines based on accessibility and community adoption. Included in our evaluation are KrissBERT (Microsoft)²⁷, ScispaCy (AI2)²², and USAGI (Observational Health Data Sciences and Informatics [OHDSI])²⁶, which remain widely used and are publicly available. In contrast, other systems, such as UMLS MetaMap (NLM, last updated in 2020)³ and ArboEL²⁸ (U. Mass, dependent on the obsolete Ab3p⁴⁷), were excluded due to maintenance and reproducibility issues. Other tools, such as SapBERT⁴⁸, QuickUMLS⁴⁹, and BLINK⁵⁰ have seen declining performance in recent evaluations and were not included in our comparison.

Human evaluation of expansion quality

To assess the quality of generated concept expansions, we conducted a structured human evaluation involving domain experts (Supplementary Fig. 2). We randomly sampled 50 biomedical mentions from the ST21pv benchmark and, for each mention, generated concept expansions using

seven different backbone LLMs. Reviewers were shown: the original biomedical mention, its generated expansion, an explanation for the generation, the context sentence, and the ground truth concept's preferred name and definition. They rated each expansion on a 5-point Likert scale (1–5) based on how reasonable, meaningful, and useful they found it for improving concept mapping. The evaluation focused on whether expansions clarified the intended meaning and improved disambiguation.

Controlled experiments on model and context effects

To investigate factors impacting workflow performance, we conduct a controlled experiment focusing on two key variables: model architecture and context window size. Specifically, to select the backbone model, we curate a set of 2000 concept mentions that do not appear in the training data to evaluate generalization on zero-shot cases. To evaluate the impact of context window sizes, we experiment with varying context window sizes, 32, 64, 128, and 256 tokens, using a subset of 50 samples. The maximum length of 256 tokens approximates the typical length of an article abstract and serves as an upper bound for the model to leverage.

Symbolic linking with or without mention expansion

To evaluate how well symbolic linking disambiguates expanded mentions and maps them to the correct CUIs, we conducted an experiment using the ST21pv dataset. This dataset is specifically curated to cover commonly used biomedical semantic types, providing a focused benchmark for evaluating disambiguation performance. We applied our expansion module to convert mentions into more explicit forms and compared the performance of symbolic linking with and without expansion.

Data availability

The data and code underlying this article are available at <https://github.com/ebmlab/MedicalConceptMapping>.

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Author contributions

Study concepts/study design, G.Z., Y.P., C.W.; manuscript drafting or manuscript revision for important intellectual content, G.Z., Y.F., F.C., C.T., G.H., P.R., Y.P., C.W.; approval of final version of the submitted manuscript, G.Z., Y.F., F.C., C.T., G.H., P.R., Y.P., C.W.; agrees to ensure any questions related to the work are appropriately resolved, G.Z., Y.F., F.C., C.T., G.H., P.R., Y.P., C.W.; literature research, G.Z., Y.P., C.W.; experimental studies, G.Z., Y.F.; data interpretation and statistical analysis, all authors; expert evaluation, F.C., C.T.

Competing interests

The authors declare no competing interests.

Additional information

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